Johan Edstedt

PhD · Computer Vision

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Education

Linköping University *Linköping, Sweden*

PhD Computer Vision *2021/03 - 2025/10*

- Advisor: Prof. Michael Felsberg
- Co-Advisor: Associate Prof. Mårten Wadenbäck
- PhD Thesis: *Towards the Next Generation of 3D Reconstruction*, DOI: `https://doi.org/10.3384/9789181181906`

Linköping University *Linköping, Sweden*

MS Engineering Physics and Electrical Engineering *2014/08 - 2020/06*

- MS Thesis: *Towards Understanding Capsule Networks*

Katholieke Universitet Leuven *Leuven, Belgium*

Erasmus Studies *2018/08 - 2019/06*

Professional Experience

2024-2024 **3D Computer Vision Intern**, Apple, Cupertino, U.S.

2023-2023 **Research Intern**, Ericsson Research, Stockholm, Sweden

2020-2021 **Research Assistant**, Linköping University, Linköping, Sweden

Publications

David Nordström, **Johan Edstedt**, Georg Bökman, Fredrik Kahl. *MuM: Multi-View Masked Image Modeling for 3D Vision*. arXiv preprint arXiv:2511.17309.

Johan Edstedt, David Nordström, Yushan Zhang, Georg Bökman, Jonathan Astermark, Viktor Larsson, Anders Heyden, Fredrik Kahl, Mårten Wadenbäck, Michael Felsberg. *RoMa v2: Harder Better Faster Denser Feature Matching*. arXiv preprint arXiv:2511.15706.

Mårten Wadenbäck, Marcus Valtonen Örnhag, **Johan Edstedt**. *Radially Distorted Homographies, Revisited* . 2026 International Conference on 3D Vision (3DV).

David Nordström, **Johan Edstedt**, Fredrik Kahl, Georg Bökman. *Stronger ViTs With Octic Equivariance*. arXiv preprint arXiv:2505.15441.

Johan Edstedt, Georg Bökman, Mårten Wadenbäck, Michael Felsberg. *DaD: Distilled Reinforcement Learning for Diverse Keypoint Detection*. arXiv preprint arXiv:2503.07347.

Ludvig Dillén, Per-Erik Forssén,**Johan Edstedt**. *FACT: Multinomial Misalignment Classification for Point Cloud Registration*. 2025 Scandinavian Conference on Image Analysis (SCIA).

Johan Edstedt, Andre Mateus, Alberto Jaenal. *ColabSfM: Collaborative Structure-from-Motion by Point Cloud Registration*. 2025 IEEE Conference on Computer Vision and Pattern Recognition (CVPR).

Johan Edstedt. *Less Biased Noise Scale Estimation for Threshold-Robust RANSAC*. 2025 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW).

Georg Bökman, **Johan Edstedt**, Michael Felsberg, Fredrik Kahl. *Affine steerers for structured keypoint description*. 2024 European Conference on Computer Vision (ECCV).

Georg Bökman, **Johan Edstedt**, Michael Felsberg, Fredrik Kahl. *Steerers: Steerers: A framework for rotation equivariant keypoint descriptors*. 2024 IEEE Conference on Computer Vision and Pattern Recognition (CVPR).

Johan Edstedt, Georg Bökman, Zhenjun Zhao. *DeDoDe v2: Analyzing and Improving the DeDoDe Keypoint Detector*. 2024 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW).

Johan Edstedt, Qiyu Sun, Georg Bökman, Mårten Wadenbäck, Michael Felsberg. *RoMa: Robust Dense Feature Matching*. 2024 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). **300 citations as of November 2025.**

Johan Edstedt, Georg Bökman, Mårten Wadenbäck, Michael Felsberg. *DeDoDe: Detect Don't Describe—Describe Don't Detect for Local Feature Matching*. 2024 International Conference on 3D Vision (3DV) **(Oral)**.

Jie Zhao, **Johan Edstedt**, Michael Felsberg, Dong Wang, Huchuan Lu. *Leveraging the Power of Data Augmentation for Transformer-based Tracking*. 2024 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV).

Yushan Zhang, **Johan Edstedt**, Bastian Wandt, Per-Erik Forssén, Maria Magnusson, Michael Felsberg. *GMSF: Global Matching Scene Flow*. 2023 Annual Conference on Neural Information Processing Systems (NeurIPS).

Johan Edstedt, Ioannis Athanasiadis, Mårten Wadenbäck, Michael Felsberg. *DKM: Dense Kernelized Matching for Geometry Estimation*. 2023 Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) **(Highlight)**.

Emil Brissman, Per-Erik Forssén, **Johan Edstedt**. *Camera Calibration Without Camera Access-A Robust Validation Technique for Extended PnP Methods* . 2023 Scandinavian Conference on Image Analysis (SCIA).

Joakim Johnander, **Johan Edstedt**, Michael Felsberg, Fahad Shahbaz Khan, Martin Danelljan. *Dense gaussian processes for few-shot segmentation* . 2022 European Conference on Computer Vision (ECCV).

Johan Edstedt, Amanda Berg, Michael Felsberg, Johan Karlsson, Francisca Benavente, Anette Novak, Gustav Grund Pihlgren. *VidHarm: A Clip Based Dataset for Harmful Content Detection* . 2022 26th International Conference on Pattern Recognition (ICPR).

Awards, Fellowships, & Grants

2023 **AISG–SLA Visual Localisation Challenge — Winning Team**, AI Singapore, SLA *$ 20,000*